14. Artificial and Natural Computers

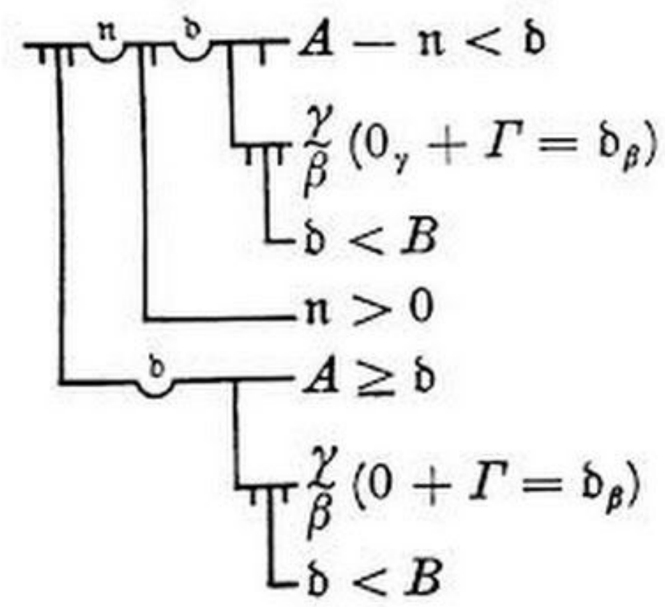
The Invention of Computers

Why create a computer? Why isn't human cognition enough? Why appeal to computation at all? Speed. Accuracy. Reliability. Scalability. Cost. Humans have developed automated computing machines for the same reason we have created any number of automated devices, from the egg beater to the steam engine.

Leibniz himself imagined a machine that would be faster and more reliable than human calculators.

> "If controversies were to arise, there would be no more need of disputation between two philosophers than between two accountants. For it would suffice to take their pencils in their hands, and say to each other: *Calculemus*—Let us calculate."
>
> — Leibniz, 1684

Meanwhile, logicians from Aristotle to Frege, attempted to formalize the laws of thought.



Frege's 1894 "Begriffsschrift" or *Concept-script.*

In the 20th century, the bridge was finally made between logic and physical systems. The promise of human-run computation was finally achieved. .

Table I. Analogue Between the Calculus of Propositions and the Symbolic Relay Analysis

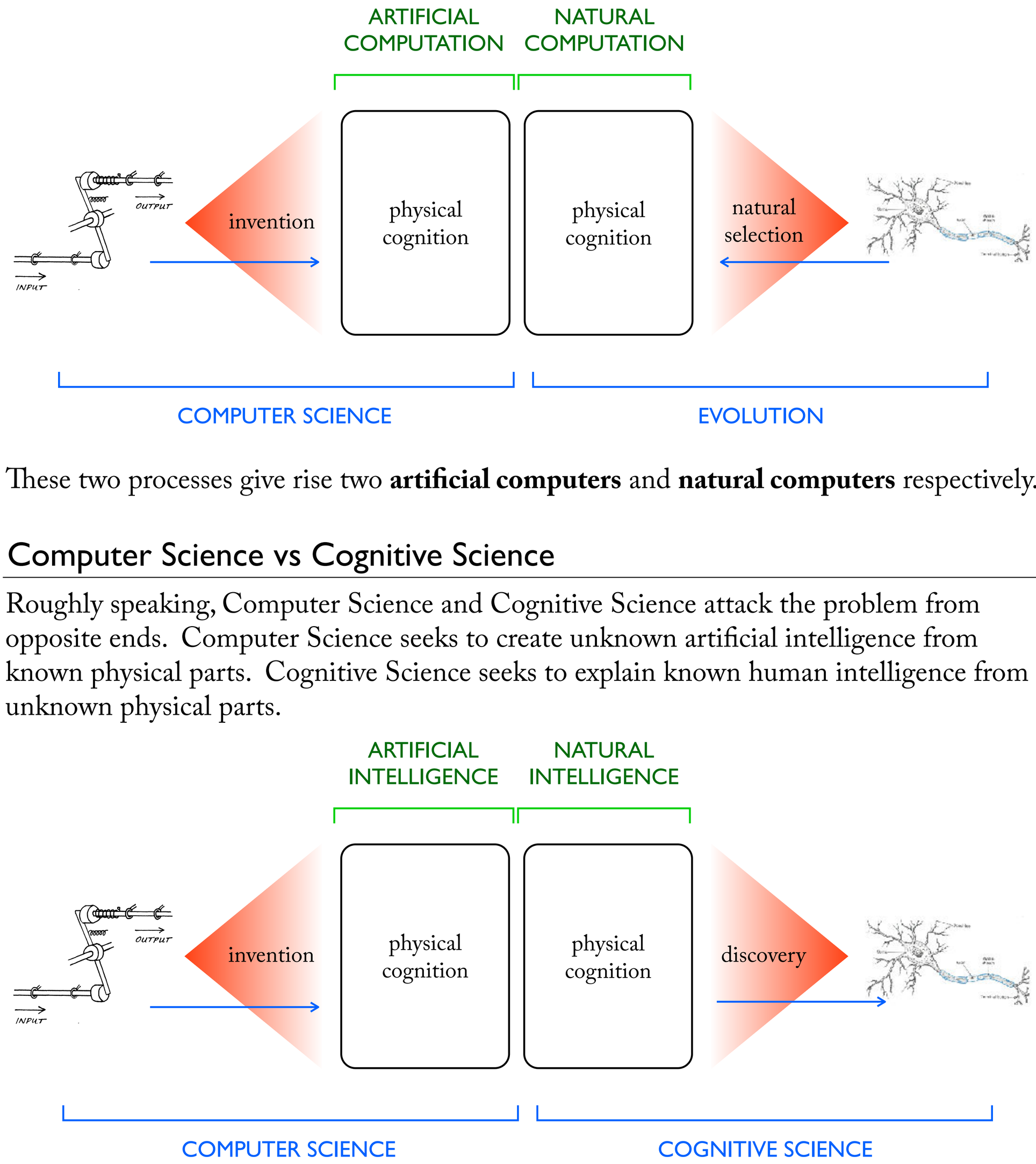
Symbol	Interpretation in Relay Circuits	Interpretation in the Calculus of Propositions
X	The circuit X	The proposition X
0	The circuit is closed	The proposition is false
1	The circuit is open	The proposition is true
$X + Y$	The series connection of circuits X and Y	The proposition which is true if either X or Y is true
$X\ Y$	The parallel connection of circuits X and Y	The proposition which is true if both X and Y are true
X'	The circuit which is open when X is closed and closed when X is open	The contradictory of proposition X
=	The circuits open and close simultaneously	Each proposition implies the other

Shannon 1938 "A symbolic analysis of relay and switching circuits"

> "Logic began as a way to understand the laws of thought. It then helped create machines that could reason according to the rules of deductive logic. Today, deductive and inductive logic are being combined to create machines that both reason and learn. What began, in Boole's words, with an investigation "concerning the nature and constitution of the human mind," could result in the creation of new minds—artificial minds—that might someday match or even exceed our own." Dixon

Computer Science vs Nature

Human Computer Science and Nature represent two processes by which physical cognition has developed.



These two processes give rise two **artificial computers** and **natural computers** respectively.

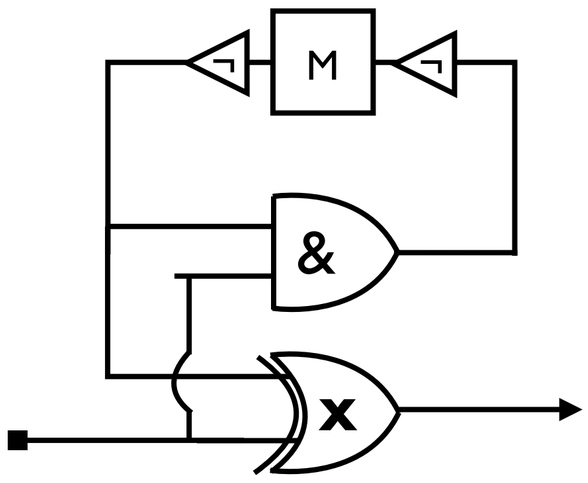
Computer Science vs Cognitive Science

Roughly speaking, Computer Science and Cognitive Science attack the problem from opposite ends. Computer Science seeks to create unknown artificial intelligence from known physical parts. Cognitive Science seeks to explain known human intelligence from unknown physical parts.

"Psychology is engineering in reverse. In forward-engineering, one designs a machine to do something; in reverse-engineering, one figures out what a machine was designed to do." Pinker (1997, 21)

Making Minds

In this class, we are straddling the divide between computer science and cognitive science, artificial and natural. We are using artificial computers as a model for what natural selection could have produced. And we are using the invented machines of computer science as a model for the discovered machines of cognitive science.





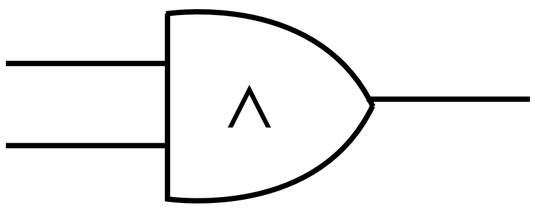
Artificial Computers

Designed. Artificial computers are designed by humans.

Assigned functions. The purpose or function of an artificial computer is what it was assigned by its human designer or user.

User interpretation. The meanings of symbols in an artificial computer are (+/-) determined by the interpretive choices of a human user.

Hardware. Modern artificial computers are built from digital logic gates in an electrical medium.



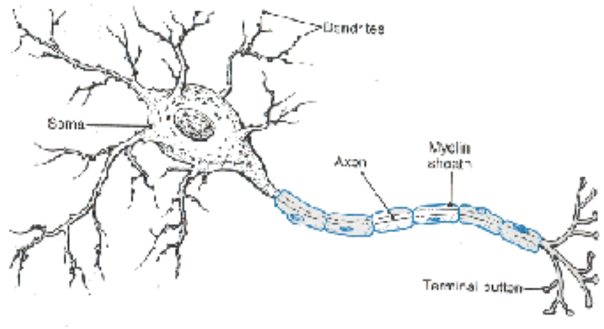
Natural Computers

Evolved. Natural computers are the product of natural selection.

Selected functions. The function of a natural computer is what it was selected to do as a result of an evolutionary process.

Natural representation. The meanings of symbols in a natural computer are determined by the organism's causal interactions with the environment.

Wetware. Natural computers are built from neurons in an electro-chemical medium.



"The mind is a system of organs of computation, designed by natural selection… The mind is what the brain does; specifically, the brain processes information, and thinking is a kind of computation."
Pinker (1997, 21)

Original intentionality. Symbols in an artificial computer are given their contents by stipulation or convention. But what gives the contents to the symbols in a natural computer? The computer concept doesn't resolve this at all, but without content there is no representation, and no computation.

> "The question naturally arises of course as to how mental symbols – mental representations – acquire their contents or meanings. The numerals of various numeral systems such as Arabic numerals and Roman numerals derive their meanings from the linguistic conventions of communities. The current along the input and output wires of an electronic computer circuit derive their meanings from our intentional stipulative assignments; as do the keys of hand-calculators and the displays on their screen. If intentionality is to be explained by appeal to mental representations, then we need an account of their meaning that makes no appeal to linguistic conventions or semantic intentions, for these presuppose intentionality… Suffice it to note that the problem of how mental representations acquire their meanings is perhaps the deepest problem that any reductive theory of intentionality faces." (McG)

From plants to informavores: plants have evolved to be responsive to information in the environment, but they do not need to store or process that information. More complex creatures take information in from the environment, processes it, and act on the basis of this processing. For these creatures, information-processing is a biological function.

Externalism: the philosophical idea that the content of your mental states depends in part on the internal state of your brain (what symbols you are processing), but also on your relationship to the external environment (how those symbols carry external information).

